ECE 310 Fall 2026
Lecture 21: DFT Properties

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Learning Objectives

After this lecture, you should be able to:

- Explain and apply DFT properties for analyzing signals in both the time-domain and DFT-domain.
- Understand the circular convolution property of the DFT and perform circular convolution of finite-length sequences.



Discrete Fourier transform (DFT)

The discrete Fourier transform of a length–N discrete-time signal is denoted by $X[k]$.

$$X[k] = \sum_{n=0}^{N-1} x[n] e^{-j\frac{2\pi k}{N}n}, \ 0 \leq k \leq N-1$$

$$x[n] = \frac{1}{N} \sum_{k=0}^{N-1} X[k] e^{j\frac{2\pi k}{N}n}, \ 0 \leq n \leq N-1$$



Periodicity of the DFT

Recall from last lecture that for a length–N signal $x[n]$ and its DFT $X[k]$:

- N-**periodicity in frequency**: → Equivalent to 2π periodicity in $X_d(\omega)$

$$X[k] = X[k + mN]$$

- N-**periodicity in time:**

$$x[n] = x[n + mN]$$



The Modulo Operation

Working with a length–N signal $x[n]$ and its DFT $X[k]$ implies *periodic extension* of $x[n]$ for $n < 0$ and $n \geq N$ (i.e. $x[n] = x[n + mN]$). Another way to denote this is with the modulo operation.

$$x[n] = x[n + mN] = x[\langle n \rangle_N]$$

Modulo Operation

The modulo operation $\langle a \rangle_N$ gives the remainder of dividing a by N. For example,

- $\langle 7 \rangle_4 = 3$
- $\langle -1 \rangle_3 = 2$
- $\langle 6 \rangle_3 = 0$



Circular Time-reversal

Reversing a periodically-extended length–N signal is given as follows:

$$x[\langle -n \rangle_N] = \begin{cases} x[0], & n = 0 \\ x[N-n], & 1 \le n \le N-1 \end{cases}$$

Ex: $x[n] = \{1, 2, 3, 4, 5\}$

$x[\langle -n \rangle_5] = \{1, 5, 4, 3, 2\}$

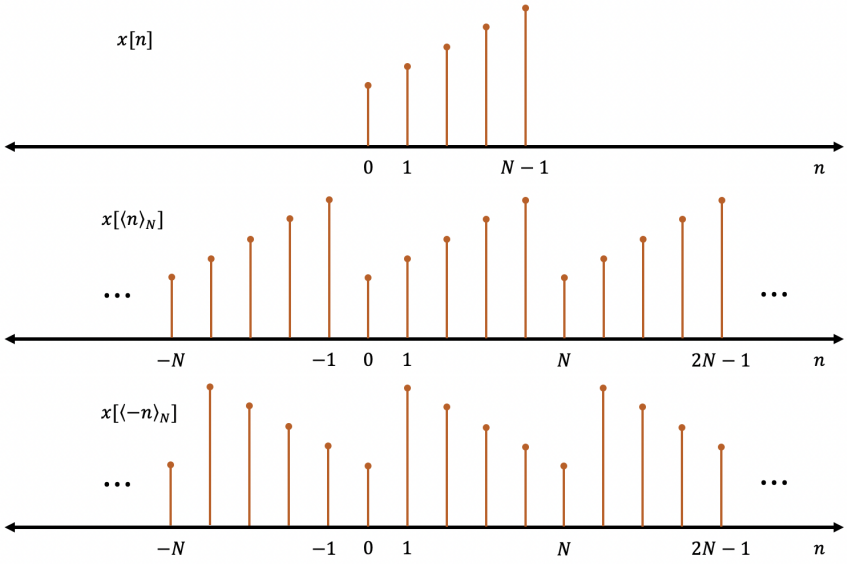
$x[\langle -n \rangle_N] = x[\langle -n+N \rangle_N]$

$x[\langle -0 \rangle_N] = x[0]$

$x[\langle -1 \rangle_N] = x[\langle N-1 \rangle_N] = x[N-1]$

$x[\langle -2 \rangle_N] = x[N-2]$

Periodic Extension



Example 1: Circular Shifting and Time-reversal

Let $x[n] = \{\underset{\uparrow}{3}, 1, 2, 4, 5\}$ be a length–5 signal. Write the length–5 sequence representing each of the following signals.

(a) $x[\langle -n \rangle_5]$

(b) $x[\langle n + 2 \rangle_5]$

(c) $x[\langle 1 - n \rangle_5]$

$$x[\langle -n \rangle_N] = \begin{cases} x[0], & n = 0 \\ x[N-n], & 1 \le n \le N-1 \end{cases}$$

(a) $x[\langle -n \rangle_5] = \{\underset{\uparrow}{3}, 5, 4, 2, 1\}$

(b) $x[\langle n+2 \rangle_5] = \{\underset{\uparrow}{2}, 4, 5, 3, 1\}$

(c) $x[\langle 1-n \rangle_5] = x[\langle -(n-1) \rangle_5] = \{\underset{\uparrow}{1}, 3, 5, 4, 2\}$

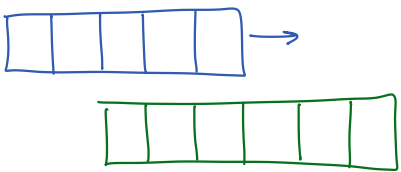
Convolution Property of the DFT

Thus far, we have seen the convolution properties of the z-transform and the DTFT.

$$x[n] * h[n] \overset{\mathcal{Z}}{\longleftrightarrow} X(z)H(z)$$
$$x[n] * h[n] \overset{\mathcal{F}}{\longleftrightarrow} X_d(\omega)H_d(\omega)$$

Do we have a similar relationship in the DFT domain?

Almost, but periodic extension complicates things



Circular Convolution

Circular Convolution Property

For length–N finite-length sequences $x[n]$ and $h[n]$,

$$x[n] \circledast h[n] \overset{\text{DFT}}{\longleftrightarrow} X[k]H[k]$$

where

$$x[n] \circledast h[n] = \sum_{m=0}^{N-1} x[m]h[\langle n-m \rangle_N] = \sum_{m=0}^{N-1} h[m]x[\langle n-m \rangle_N]$$

- Circular convolution is performed between equal-length sequences.
- If one sequence is shorter, we zero-pad this sequence to match the length of the longer sequence.
- We sometimes write circular convolution with $\circledast_N$ to denote that both sequences are padded to length–N as necessary.

Computing Circular Convolution: Table Method

Let

$$x[n] = \{3, 1, 0, 3, 1, 1\}$$
$$h[n] = \{-4, -1, -7, 5, 5, 1\}.$$

$$x[n] \circledast_6 h[n] = \sum_{m=0}^{5} x[m]\, h[\langle n-m \rangle_6]$$

We would like to compute $y[n] = x[n] \circledast h[n]$.

	m	0	1	2	3	4	5	
	$x[m]$	3	1	0	3	1	1	$y[n]$
$n=0 \rightarrow$	$h[\langle -m \rangle_N]$	-4	1	5	5	-7	-1	$y[0] = -4$
$n=1 \rightarrow$	$h[\langle 1-m \rangle_N]$	-1	-4	1	5	5	-7	6
$n=2 \rightarrow$	$h[\langle 2-m \rangle_N]$	-7	-1	-4	1	5	5	-9
	$h[\langle 3-m \rangle_N]$	5	-7	-1	-4	1	5	2
	$h[\langle 4-m \rangle_N]$	5	5	-7	-1	-4	1	14
	$h[\langle 5-m \rangle_N]$	1	5	5	-7	-1	-4	-18

$$y[n] = \{-4, 6, -9, 2, 14, -18\}$$

Computing Circular Convolution: Matrix Method

$$x[n] = \{3, 1, 0, 3, 1, 1\}$$
$$h[n] = \{-4, -1, -7, 5, 5, 1\}.$$

We can compute $y[n] = x[n] \circledast h[n]$ instead by matrix-vector multiplication where $y = \tilde{\mathbf{H}}x = \tilde{\mathbf{X}}h$ for circular convolution matrices $\tilde{\mathbf{H}}$ and $\tilde{\mathbf{X}}$.

$$y = \begin{bmatrix} -4 & 1 & 5 & 5 & -7 & -1 \\ -1 & -4 & 1 & 5 & 5 & -7 \\ -7 & -1 & -4 & 1 & 5 & 5 \\ 5 & -7 & -1 & -4 & 1 & 5 \\ 5 & 5 & -7 & -1 & -4 & 1 \\ 1 & 5 & 5 & -7 & -1 & -4 \end{bmatrix} \begin{bmatrix} 3 \\ 1 \\ 0 \\ 3 \\ 1 \\ 1 \end{bmatrix} = \{-4, 6, -9, 2, 14, -18\}$$

$\tilde{H}$

Computing Circular Convolution: Matrix Method

$$x[n] = \{3, 1, 0, 3, 1, 1\}$$
$$h[n] = \{-4, -1, -7, 5, 5, 1\}.$$

We can compute $y[n] = x[n] \circledast h[n]$ instead by matrix-vector multiplication where $y = \tilde{\mathbf{H}}x = \tilde{\mathbf{X}}h$ for circular convolution matrices $\tilde{\mathbf{H}}$ and $\tilde{\mathbf{X}}$.

$$y = \begin{bmatrix} 3 & 1 & 1 & 3 & 0 & 1 \\ 1 & 3 & 1 & 1 & 3 & 0 \\ 0 & 1 & 3 & 1 & 1 & 3 \\ 3 & 0 & 1 & 3 & 1 & 1 \\ 1 & 3 & 0 & 1 & 3 & 1 \\ 1 & 1 & 3 & 0 & 1 & 3 \end{bmatrix} \begin{bmatrix} -4 \\ -1 \\ -7 \\ 5 \\ 5 \\ 1 \end{bmatrix} = \{-4, 6, -9, 2, 14, -18\}$$

$\tilde{X}$ h

Example 2: Computing Circular Convolution

Compute the circular convolution of $x[n]$ and $h[n]$ below.

$$x[n] = \{\underset{\uparrow}{2}, 0, 1, 3, 0\}$$

$$h[n] = \{\underset{\uparrow}{4}, 1, 0, 2, 2\}$$

$$y = \tilde{H}x = \begin{bmatrix} 4 & 2 & 2 & 0 & 1 \\ 1 & 4 & 2 & 2 & 0 \\ 0 & 1 & 4 & 2 & 2 \\ 2 & 0 & 1 & 4 & 2 \\ 2 & 2 & 0 & 1 & 4 \end{bmatrix} \begin{bmatrix} 2 \\ 0 \\ 1 \\ 3 \\ 0 \end{bmatrix}$$

$$= \begin{bmatrix} 10 \\ 10 \\ 10 \\ 21 \\ 7 \end{bmatrix}$$

$$y[n] = \{\underset{\uparrow}{10}, 10, 10, 17, 7\}$$

Summary of DFT Properties

Many other DFT properties follow from DTFT properties except with **circularity**, e.g. circular time shifting, circular time reversal.

Property	Length-N $x[n]$	DFT $X[k]$
Linearity	$ax_1[n] + bx_2[n]$	$aX_1[k] + bX_2[k]$
Circular time shifting	$x[\langle n-m\rangle_N]$	$e^{-j\frac{2\pi mk}{N}}X[k]$
Circular frequency shifting	$e^{j\frac{2\pi mn}{N}}x[n]$	$X[\langle k-m\rangle_N]$
Circular modulation	$\cos\left(\frac{2\pi m}{N}n\right)x[n]$	$\frac{1}{2}X[\langle k+m\rangle_N] + \frac{1}{2}X[\langle k-m\rangle_N]$
Circular time reversal	$x[\langle -n\rangle_N]$	$X[\langle -k\rangle_N]$
Conjugation	$x^*[n]$	$X^*[\langle -k\rangle_N]$
Duality	$X[n]$	$Nx[\langle -k\rangle_N]$
Circular convolution	$x[n] \circledast h[n]$	$X[k]H[k]$
Windowing	$x[n]w[n]$	$\frac{1}{N}X[k] \circledast W[k]$
Parseval's theorem	$\sum_{n=0}^{N-1} x[n]y^*[n]$	$\frac{1}{N}\sum_{k=0}^{N-1} X[k]Y^*[k]$
Parseval's relation	$\sum_{n=0}^{N-1} \lvert x[n]\rvert^2$	$\frac{1}{N}\sum_{k=0}^{N-1} \lvert X[k]\rvert^2$



Example 3: Applying DFT Properties

Let $x[n] = \{1, 2, 3, 4, 5, 6\}$ be a length–6 signal with DFT $X[k]$ given below:

$n=0$

$$X[k] = \{X_0, X_1, X_2, X_3, X_4, X_5\}.$$

$k=0$

Another signal $y[n]$ has DFT $Y[k]$:

$$Y[k] = \{X_2, -X_3, X_4, -X_5, X_0, -X_1\}$$

$k=0$

Determine $y[n]$.

$Y[k] = X[\langle k-4 \rangle_6] \cdot (-1)^k$ → $Y[k] = V[k](-1)^k = V[k] e^{-j\frac{2\pi \cdot 3}{6}k}$

$e^{j\frac{2\pi m}{N}n} x[n] \overset{DFT}{\longleftrightarrow} X[\langle k-m \rangle_N]$

$x[\langle n-m \rangle_N] \overset{DFT}{\longleftrightarrow} e^{-j\frac{2\pi k}{N}m} X[k]$

$(-1)^k = e^{-j\pi k}$

$y[n] = v[\langle n-3 \rangle_6]$ (Circular time-shifting)

→ $V[k] = X[\langle k-4 \rangle_6]$ → $y[n] = e^{j\frac{4\pi}{3}\langle n-3 \rangle_6} x[\langle n-3 \rangle_6]$

$v[n] = e^{j\frac{2\pi \cdot 4}{6}n} x[n]$

$= e^{j\frac{4\pi}{3}n} x[n]$

$y[n] = \{4, 5e^{j\frac{16\pi}{3}}, 6e^{j\frac{20\pi}{3}}, 1, 2e^{j\frac{4\pi}{3}}, 3e^{j\frac{8\pi}{3}}\}$

↑ $n=0$